MIL-T-47060(MI)
10 May 1974
SUPERSEDING
MIS 14158A
29 December 1965

# MILITARY SPECIFICATION

# TOLUENE - 2,4 DIAMINE

This specification is approved for use by all departments and agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers technical grade toluene-2,4 diamine.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-P-14232 | Parts, Equipment and Tolls for Army Materiel, Packaging and Packing of |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.



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American Society for Testing and Materials

| | |
|---|---|
| ASTM D 1303 | Total Chlorine in Vinyl Chloride Polymer and Copolymers |
| ASTM D 128 | Method of Analysis of Lubricating Grease |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specification and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), the preproduction sample shall meet the requirements of this specification.

3.2 Chemical and physical analysis. The chemical and physical analysis of the material shall be as specified in Table I.

Table I. Chemical and physical analysis

| Characteristics | Minimum | Maximum |
|---|---|---|
| Toluene-2,4 diamine, percent | 99.0 | ---- |
| Water insolubles, percent | ---- | 0.10 |
| Chlorine, percent | ---- | 0.10 |
| Melting point, degree C | 97.0 | ---- |
| Moisture, percent | ---- | 1.00 |
| Ash, percent | ---- | 0.20 |
| Acid insolubles, percent | ---- | 5.00 |

3.3 Workmanship. The toluene-2,4 diamine shall be uniform in quality, free from foreign materials, and manufactured under conditions and procedures standard to the industry.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptance inspection. Acceptance inspection shall, at the option of the Government, be performed either by the supplier or by a Government representative. In either case, responsibility for final acceptance rests with the Government representative.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of toluene-2,4 diamine. The preproduction sample shall be subjected to all the examinations and tests specified herein. Unless otherwise specified, the Government will perform the examinations and tests for preproduction sample acceptance at the contractor's plant. Preproduction samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Lot. A lot shall consist of material produced at one plant with no change in formulation or process. If manufacture is by batch process, each batch shall constitute a lot. A batch shall be as defined in 6.3.

4.4 Acceptance sampling. The number of containers to be chosen at random for acceptance sampling shall be equal to the square root of the total number of containers in the lot. If the number thus obtained is not a whole number, the number of containers to be sampled shall be increased to the next higher whole number. In no case, however, shall the number of containers to be sampled be less than seven (unless there are less than seven containers in the lot; in this case, each container shall be sampled).

4.4.1 Primary sample. From each selected container, a sample shall be taken from three or more places throughout the container. The total weight of the sample taken from each container shall weigh at least 50 grams (g). Each sample thus taken shall be mixed thoroughly,



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placed in a clean, dry container, and labeled to identify the material name, original container designation, contract number, and lot number.

4.4.2 Composite sample. Each primary sample shall be subdivided to prepare a composite sample (not in excess of 500g). Primary material not used shall be returned to the primary sample container. After mixing the composite sample thoroughly, the composite sample shall be placed in a clean, dry container and sealed. The composite sample shall be identified with the material name, container designation, contract number, and lot number. All specified chemical and physical tests shall be made on this composite sample representing the lot. Failure of the composite sample to pass all of the tests herein shall result in rejection of the lot represented.

4.5 Classification of tests. Inspection and testing of toluene-2,4 diamine shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.6 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all examinations and tests specified herein.

4.7 Quality conformance tests. Quality conformance tests for acceptance of the toluene-2,4 diamine shall consist of the following tests:

| Characteristics | Tests |
|---|---|
| Toluene-2,4 diamine | 4.8.1 |
| Water insolubles | 4.8.2 |
| Chlorine | 4.8.3 |
| Melting point | 4.8.4 |
| Moisture | 4.8.5 |
| Ash | 4.8.6 |
| Acid insolubles | 4.8.7 |

4.8 Tests. The following procedures shall be used to determine that the requirements of this specification have been met. Any changes in the test procedures employed shall be subject to prior approval of



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the procuring activity. Unless otherwise specified, all tests shall be run in duplicate. The average of the two results shall be taken as the test result.

4.8.1 Toluene-2,4 diamine.

4.8.1.1 Principle. The method for the determination of the purity of toluene-2,4 diamine depends upon a titration with diazo p-chlorobenzene solution in an acetic acid medium. This method shall be used for analysis of technical grade toluene-2,4 diamine.

4.8.1.2 Diazotized p-chloroaniline (diazo p-chlorobenzene) standard solution.

a. Reagents.

1. Para-chloroaniline, technical grade.

2. Concentrated hydrochloric acid.

3. Sulfamic acid, technical grade.

4. Potassium bromide crystals, analytical grade.

5. Potassium bromate-bromide ($KBrO_3$-KBr) solution 1/8 normal (N/8), standardized.

b. Procedures.

1. Preparation of 0.5 N p-chloroaniline hydrochloride solution. To prepare one liter of 0.5 N solution dissolve approximately 63.8 g of p-chloroaniline in 500 milliliters (ml) of distilled water containing 100 ml of concentrated hydrochloric acid. Heat, if necessary, to complete the dissolution. Add a small quantity of Nuchar, mix well, and filter. Pour the filtrate into a one-liter volumetric flask and dilute to volume. (After standardization, store in an amber-color flask.)

2. Standardization of the 0.5 N p-chloroaniline hydrochloride solution. Pipet 50 ml of the p-chloroaniline solution into a one-liter volumetric flask and dilute to volume. Mix thoroughly and pipet a 50-ml aliquet into a 400-ml beaker. Add 200ml of distilled water



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and 15 ml of hydrochloric acid. Titrate with N/8 $KBrO_3$-KBr solution, adding the $KBrO_3$-KBr solution drop-wise until a spot test with potassium (or cadmium) iodide-starch papers show a faint distinct spot for 2 minutes. Make a blank determination omitting only the p-chloroaniline.

$$\text{Normality of p-chloroaniline hydrochloride solution} \text{ equals } \frac{(A \text{ minus } B) \text{ times } N}{2.5 \text{ times } 4}$$

Where: A equals volume of $KBrO_3$-KBr used to titrate sample, ml

B equals volume of $KBrO_3$-KBr used to titrate blank, ml

N equals normality of $KBrO_3$-KBr

3. Preparation of N/40 solution of diazo p-chlorobenzene hydrochloride solution. Pipet 50 ml of 0.5 n p-chloroaniline solution into a 600-ml beaker. Add 300 ml of distilled water, 10 ml of concentrated hydrochloric acid, and about 10 g of potassium bromide. Then add, by rapid pouring, a measured amount of 0.5 N sodium nitrite solution required to diazotize the p-chloroaniline plus 1.0 ml in excess. Stir thoroughly, let stand 10 to 15 minutes, then add saturated sulfamic acid solution drop-wise until no excess sodium nitrite is indicated when spotted on potassium (or cadmium) iodide-starch indicator papers. Transfer the diazo solution into an amber-colored, 1000-ml volumetric flask and dilute to volume.

$$\text{Normality of the diazo p-chlorobenzene solution} \text{ equals } \frac{\text{Normality of p-chloroaniline hydrochloride solution}}{20}$$

4.8.1.3 Procedure. Weigh out a sample equivalent to 0.25 to 0.27 g of 100-percent toluene-2,4 diamine, transfer to a 600-ml beaker, and dissolve in about 75 ml of distilled water. Add 5 ml of glacial acetic acid, 20 g of sodium acetate free from carbonates, and 20 g of sodium



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chloride. Make up to 300 ml with water and titrate at room temperature, about 25 degrees Celsius (C) (77 degrees Fahrenheit (F)) with N/40 diazo p-chlorobenzene solution. Follow the titration for the disappearance of toluene-2,4 diamine by spot-testing against the diazo p-chlorobenzene solution and for excess of diazo solution by spot-testing (see 4.8.1.4) against a saturated solution of toluene-2,4 diamine. The reaction is complete when no line is formed with the diazo spot and a positive endpoint against toluene-2,4 diamine persists for 3 minutes. The diazo p-chlorobenzene solution should be added in 0.1-ml portions near the endpoint. During the titration, lines may be obtained with both indicators. This indicates that the temperature of the sample is too low or the diazo is being added to rapidly. In such cases, make the necessary adjustments and do not add any diazo solution until one of the spot tests has disappeared.

$$\text{Percent 2,4 toluene diamine equals } \frac{A \text{ times } B \text{ times } 12.22}{C}$$

Where: A equals volume of diazo solution used in titration, ml

B equals normality of diazo solution

C equals weight of sample, g

4.8.1.4 Notes. Follow the reaction by means of what is called a "negative" spot test. Make a line about 1 inch (in.) long upon a strip of spot test paper (Reeve Angel Filter Paper) with a stirring rod moistened with the titrated solution and allow it to stand about 1/4 minute or until a colorless rim has formed around the line. Then moisten the end of another stirring rod with diazo solution and make a parallel line on the spot test paper about 1/2 in. from the first line. The rim of this line should spread into the first line. As long as the uncoupled sample remains in solution, color will be produced at the point of contact of the two lines. When the coupling is practically complete, no color will be formed, and testing for the endpoint is commenced. The endpoint or "positive" spot test is detected by means of a saturated solution of toluene-2,4 diamine in water. Make a line about 1 in. long upon a spot test paper with a stirring rod moistened with the toluene-2,4 diamine solution. Make a similar and parallel line with the solution being titrated about 1/2 in. from the indicator line. The colorless rim of the dye spot will then spread into the toluene-2,4 diamine, and a trace of color will form at the junction, indicating the completion of the coupling and the presence of slight excess of diazo solution.

4.8.1.5 Acceptance criteria. For the lot represented to pass the toluene-2,4 diamine assay test, the value obtained for the percent toluene-2,4 diamine shall be not less than the value specified in 3.2.



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4.8.2 Water insolubles.

4.8.2.1 Procedure. Dissolve approximately 20 g of sample (weighed to the nearest 0.001 g) in one liter of distilled water. Filter the solution by suction through a fritted glass funnel.

$$\text{Percent water insolubles equals } \frac{B \text{ times } 100}{A}$$

Where: A equals weight of sample dissolved in water, g

B equals weight of residue collected on filter, g

4.8.2.2 Acceptance criteria. For the lot represented to pass the water-insolubles test, the value obtained for the percent water insolubles shall be no greater than the value specified in 3.2.

4.8.3 Chlorine. Determine the amount of chlorine utilizing the procedures given by ASTM D 1303, except use ferric alum indicator in place of a ferric nitrate indicator, the electrical ignition of the Parr bomb in place of ignition by use of a Bunsen burner.

4.8.3.1 Acceptance criteria. For the lot represented to pass the chlorine test, the value obtained for the percent chlorine shall be no greater than the value specified in 3.2.

4.8.4 Melting point.

4.8.4.1 Apparatus. Melting Point Apparatus, Fisher-Hohns, or equal.

4.8.4.2 Procedure. Place a few crystals of toluene-2,4 diamine on an 18-millimeter-diameter micro-cover glass. Cover the crystals with an identical cover glass and transfer to the cleaned heating stage of a Fisher-Hohns melting point apparatus, or equal (4.8.4.1). Turn on the apparatus, bring the temperature to 90 degrees C (194 degrees F), and hold the apparatus at that temperature for at least 5 minutes. Then adjust the reading of the current-input dial to give a temperature rise rate of 0.8 to 1.0 degree C (1.4 to 1.8 degrees F) per minute as indicated the attached thermometer. With the aid of the illuminating and magnifying unit, observe the melting point in degrees C at which the last crystal of toluene-2,4 diamine liquefies.

4.8.4.3 Acceptance criteria. For the lot represented to pass the melting-point test, the value obtained for the melting point in degrees C shall be not less than the value specified in 3.2.

4.8.5 Moisture.

4.8.5.1 Apparatus. The apparatus used for determination of moisture content of the sample shall be an Aquameter, Model KF-2 or KF-3, Beckman Instruments, Inc., Fullerton, California, or an approved



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equivalent. The aquameter shall be prepared for operation as described in the technical manual furnished by the manufacturer (Beckman Instruments, Inc.). Use of an alternate equivalent item of equipment approved by the procuring activity will necessitate use of the specific technical manual prepared by the manufacturer.

4.8.5.2 Reagents.

a. Karl Fischer reagent. Karl Fischer reagent must have a strength such that each milliliter of Karl Fischer reagent corresponds to 0.0014 to 0.0023 g of water. Dilute 750 ml of commercially available stabilized Karl Fischer reagent (with water equivalent of 0.005 to 0.007 g/ml) to 2000 ml with absolute methanol (0.1 percent water, maximum). Mix well and allow to stand overnight before use. Determine the water equivalent (A) of this solution as follows:

1. Use sodium tartrate dihydrate ($Na_2C_4H_4O_6 2H_2O$) as a primary standard (with a water content of 15.66 percent) for standardizing Karl Fischer reagent. If the water content value is in question, it may be determined by heating some of the salt at 150 degrees C (302 degrees F) for 3 hours. Should the value (as determined) differ from the theoretical value of 15.66, then the experimental value shall be used in the determination of water equivalent (A) of the Karl Fischer reagent; instead of the 15.66 in the formula below, the factor should be 10P, where P is percentage moisture (as determined). Rapidly transfer 0.090 to 0.110 g (weighed to the nearest 0.0001 g) of reagent-grade $Na_2C_4H_4O_6 2H_2O$ to the titration vessel.

2. Titrate to an endpoint in the same manner as with the sample (see 4.8.5.3).

3. Repeat the standardization procedure until three successive results agree within five parts per thousand.

4. If the indicated water equivalent (A) of the Karl Fischer reagent is less than 0.0014 g of water per ml of Karl Fischer reagent, it may be due to the presence of too much water in the absolute methanol used.



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In this case, distill the methanol from metallic calcium or calcium hydride. Passing the methanol through a column of Molecular Sieves, Type 4A, or equal, may also reduce the water content of the methanol sufficiently.

Water equivalent (A) equals $\frac{0.1566\ W}{V}$

Where: A equals water equivalent of the Karl Fischer Reagent, g/ml

W equals weight of $Na_2C_4H_4O_6 2H_2O$ taken, g

V equals volume of Karl Fischer reagent used, ml

b. Water-in-methanol solution. The water-in-methanol solution should contain 0.0015 to 0.0020 g of water per ml of solution. A good grade of commercial absolute methanol contains about 0.0010 g water per ml of methanol. Water content can be adjusted by adding 1.0 g of water to 1000 ml of the water-in-methanol solution in terms of Karl Fischer reagent as follows:

1. Put about 50 ml of the anhydrous methanol used in 4.8.5.2a into the titration beaker of the Aquameter. Add a slight excess of Karl Fischer reagent (4.8.5.2a), then back-titrate with water-in-methanol solution (4.8.5.2b). Then run in an additional 5 to 8 ml of Karl Fischer reagent, read to the nearest 0.01 ml, and again back-titrate with water-in-methanol solution (read to the nearest 0.01 ml). Repeat the addition and back-titrating steps twice more to provide triplicate determinations of the equivalency ratio. Calculate the ratio (B) of Karl Fischer reagent to that of the water-in-methanol solution. The range of the ratios calculated from the three titrations shall be not greater than 0.04. If the range exceeds 0.04, continue making titrations until three ratios are obtained whose range does not exceed 0.04. Then determine the average ratio from all the ratios which have been obtained.



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4.8.5.3 Procedure. Determine the moisture content of the sample by the Karl Fischer method using a direct-titration technique. Introduce to the titration beaker, through the opening in the diaphragm, approximately 10 g of sample weighed to the nearest 0.001 g and a mixture of 150 ml of anhydrous methanol with 50 ml of glacial acetic acid. Both materials are to be reagent grade. Close the opening, start the stirrer, and press the titrate button to titrate with Karl Fischer reagent (4.8.3.2a). When the indicator light glows, read the Karl Fischer buret to the nearest 0.01 ml. Where necessary, water-in-methanol solution (4.8.3.2b) may be used to back-titrate.

$$\text{Percent moisture equals } \frac{100A\ (V_{KF} \text{ minus } BV_{WM})}{W}$$

Where: A equals weight of water equivalent to 1.00 ml of Karl Fischer reagent, g/ml

$V_{KF}$ equals volume of Karl Fischer reagent titrant used, ml

$V_{WM}$ equals volume of water-in-methanol solution titrant used, ml

B Equals ratio of Karl Fischer reagent to that of water-in-methanol solution, ml/ml

W equals weight of sample taken, g

4.8.5.4 Acceptance criteria. Fpr the lot represented to pass the moisture test, the value obtained for percent moisture shall be not greater than the value specified in 3.2.

4.8.6 Ash. Determine the amount of ash utilizing the procedure given by ASTM D 128.

4.8.6.1 Acceptance criteria. For the lot represented to pass the ash test, the value obtained for percent ash shall be not greater than the value specified in 3.2.

4.8.7 Acid insolubles.

4.8.7.1 Procedure. Dissolve approximaterly 1 g of sample (weighed to the nearest 0.0001 g) in 100 g of 5-percent hydrochloric acid at room temperatures. Filter the solution by suction through a fritted glass funnel.



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$$\text{Percent acid insolubles equals } \frac{B \text{ times } 100}{A}$$

Where: A equals weight of sample dissolved in hydrochloric acid, g

B equals weight of residue collected on filter, g

4.8.7.2 Acceptance criteria. For the lot represented to pass the acid-insolubles test, the value obtained for percent acid insolubles shall be not greater than the value specified in 3.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be in drums, barrels, or cans the size as specified in the contract.

5.1.1 Preservation, packaging, packing, and marking. Unless otherwise specified by the contract or purchase order, preservation, packaging, packing, and marking requirements for the specified levels of protection shall be in accordance with MIL-P-14232.

5.2 Marking. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129. If the material has toxic characteristics, the proper toxicity precautions shall be listed on the package.

6. NOTES

6.1 Intended use. Toluene-2,4 diamine described in this specification is intended for use as a stablizer in ammonium-nitrate-based solid propellants.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of specification.

b. Whether a preproduction sample is required (see 3.1).

c. Type and size of shipping container (see 5.1.1).

6.3 Definition.

6.3.1 Batch. A batch is defined as that quantity of material which has been subjected to one or more chemical or physical processes (or combinations thereof) intended to produce a desired product having substantially uniform characteristics.



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6.4 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-14158A, dated 29 December 1965.

Custodian:
ARMY-MI

Preparing Activity:
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